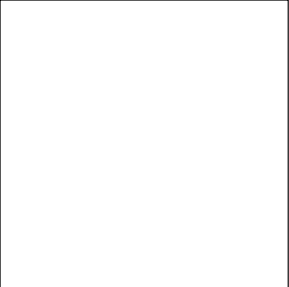
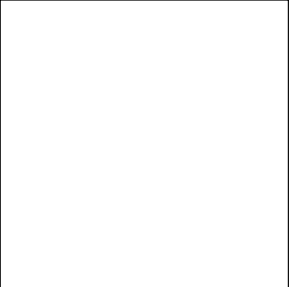


Microcontroller



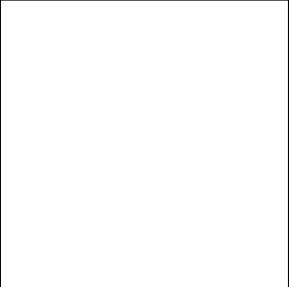
File: microcontroller.kicad_sch

Sensors



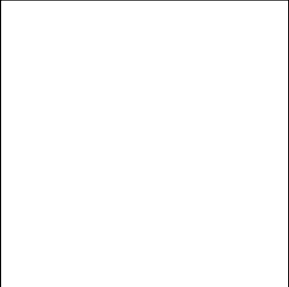
File: sensors.kicad_sch

RF

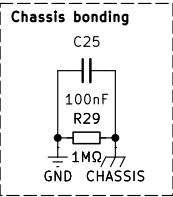
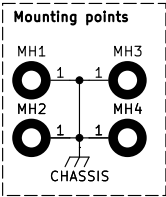


File: rf.kicad_sch

Power



File: power.kicad_sch



Organization: Lapin Rocketry
Designer: Gianluca Gaiardi
Reviewer: Gianluca Gaiardi
License: CERN-OHL-S
URL:

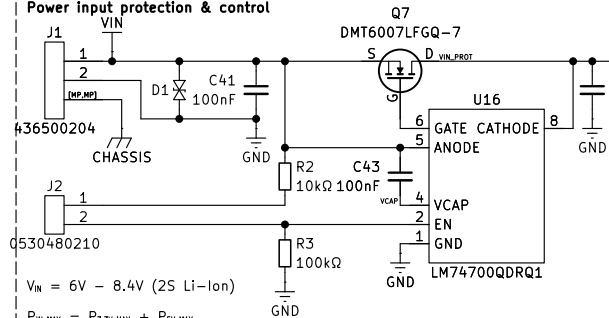


Sheet: /
File: flight_computer.kicad_sch

Flight Computer

Size: A4	Date: 19/04/2026	Revision: A
KiCad E.D.A. 10.0.4		Sheet 1 of 5

Power input protection & control

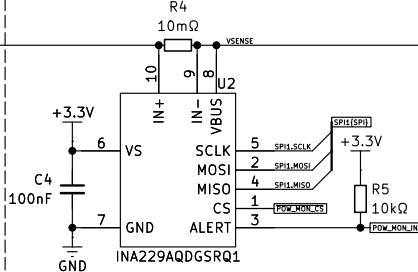


$$V_{IN} = 6V - 8.4V \text{ (2S Li-Ion)}$$

$$P_{IN\text{ MAX}} = P_{3.3V\text{ MAX}} + P_{5V\text{ MAX}} + P_{SENSE\text{ MAX}} + P_{PYRO\text{ CH MAX}} = 11.65W + 16.67W + 0.34W + 6.22W = 34.88W$$

$$I_{IN\text{ MAX}} = P_{IN\text{ MAX}} / V_{IN\text{ MIN}} = 34.77W / 6V = 5.81A$$

Power monitoring



$$V_{SENSE\text{ MAX}} = 163.84mV$$

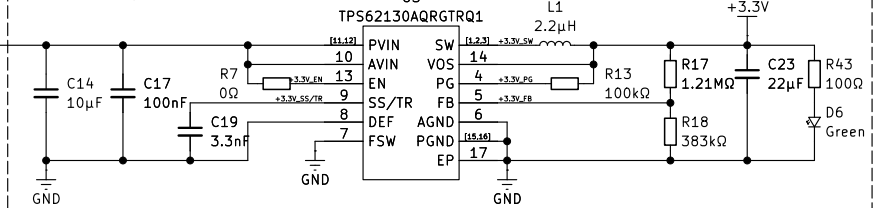
$$R_{SHUNT\text{ MAX}} = V_{SENSE\text{ MAX}} / I_{MAX} = 163.84mV / (34.54W / 6V) = 28.46mΩ = 10mΩ$$

$$P_{MAX} = R_{SHUNT} \times I_{MAX}^2 = 10mΩ \times 5.79A^2 = 0.34W$$

Pyrotechnic channels switch

0530480210

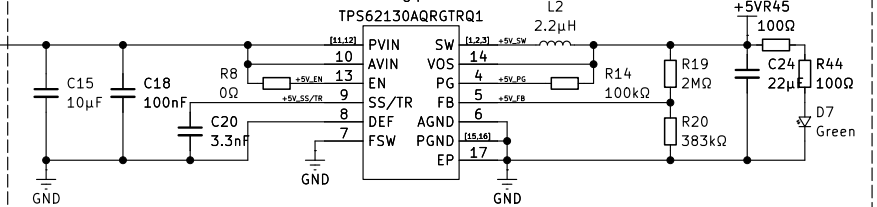
3.3V DC/DC Regulator



$$V_{OUT} = ((R_{TOP} / R_{BOTTOM}) + 1) \times 0.8V_D = (V_{IN} - V_F) / R = ((1.21MΩ / 383kΩ) + 1) \times 0.8V_{3.3V} - 2.2V / 100Ω = 11mA$$

$$P_{IN\text{ MAX}} = (P_{3.3V} / \eta_{3.3V @ 3A}) = ((3.3V \times 3A) / 0.85) = 11.65W$$

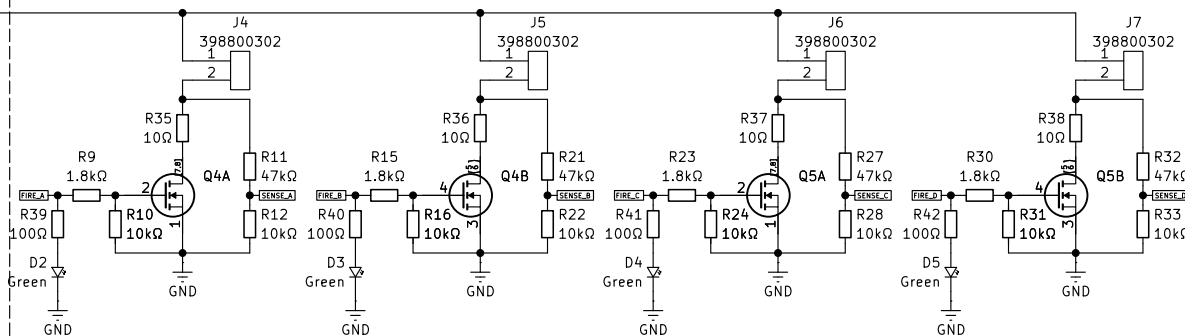
5V DC/DC regulator



$$V_{OUT} = ((R_{TOP} / R_{BOTTOM}) + 1) \times 0.8V_D = (V_{IN} - V_F) / R = ((2MΩ / 383kΩ) + 1) \times 0.8V = 5V - 2.2V / 200Ω = 14mA$$

$$P_{IN\text{ MAX}} = P_{5V} / \eta_{5V @ 3A} = ((5V \times 3A) / 0.9) = 16.67W$$

Pyrotechnic channel A,B,C,D



$$R_{LIMIT} = 10Ω$$

$$t = 100ms$$

$$I_{SENSE\text{ MAX}} = V_{MAX} / R_{TOT} = 8.4V / (47kΩ + 10kΩ) = 0.15mA$$

$$I_{MAX\text{ CH}} = V_{MAX} / R_{LIMIT} = 8.4V / (1.2Ω + 10Ω) = 0.84mA$$

$$I_{MIN\text{ CH}} = V_{MIN} / R_{LIMIT} = 6V / (1.8Ω + 10Ω) = 0.56mA$$

$$P_{LIMIT\text{ MAX}} = R_{LIMIT} \times I_{MAX}^2 = 10Ω \times 0.74A^2 = 5.48W$$

$$P_{PYRO\text{ CH MAX}} = V_{MAX} \times I_{MAX} = 8.4V \times 0.74A = 6.22W$$

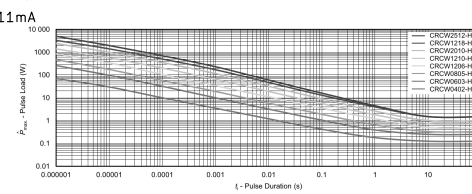
$$I_{LED} = (V_{IN} - V_F) / R = 3.3V - 2.2V / 100Ω = 11mA$$

$$I_{SENSE\text{ MIN}} = V_{MIN} / R_{TOT} = 6V / (47kΩ + 10kΩ) = 0.10mA$$

$$V_{SENSE\text{ MAX}} = V_{MAX} \times (R_{BOTTOM} / (R_{TOP} + R_{BOTTOM})) = 8.4V \times (10kΩ / (47kΩ + 10kΩ)) = 1.47V$$

$$V_{SENSE\text{ MIN}} = V_{MIN} \times (R_{BOTTOM} / (R_{TOP} + R_{BOTTOM})) = 6V \times (10kΩ / (47kΩ + 10kΩ)) = 1.05V$$

Single Pulse



Maximum pulse load, single pulse; applicable if $P \rightarrow 0$ and $n < 1000$ and $Q \leq Q_{max}$; for permissible resistance change equivalent to 8000 h operation



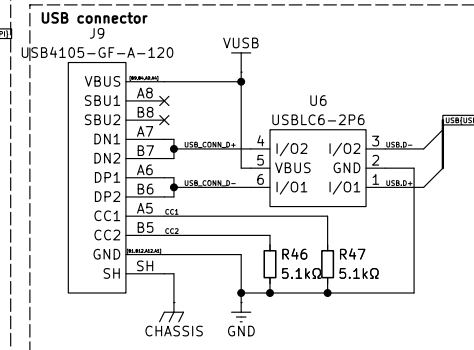
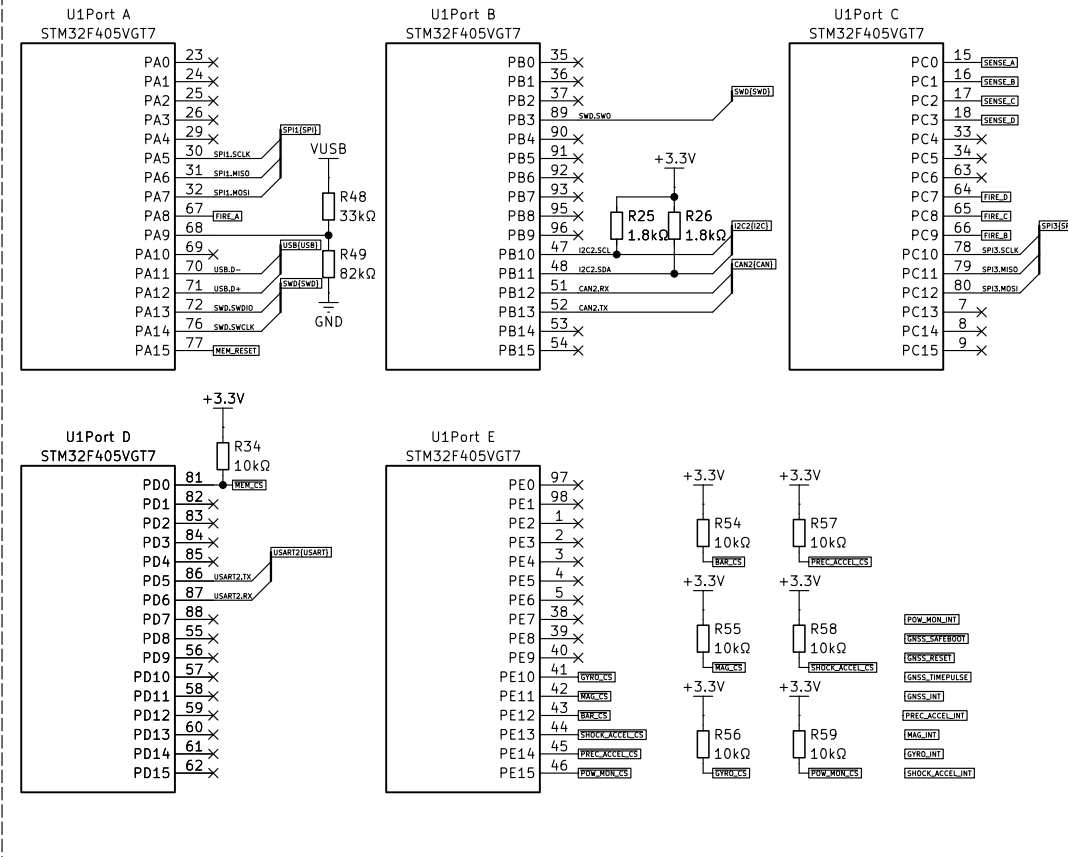
Organization: Lapin Rocketry
Designer: Gianluca Gaiardi
Reviewer: Gianluca Gaiardi
License: CERN-OHL-S
URL:



Sheet: /Power/
File: power.kicad_sch

Size: A4
Date:
KiCad E.D.A. 10.0.4

Revision:
Sheet 2 of 5



The diagram shows the wiring for the Programming/debug connector (J8). The connector has 8 pins, numbered 1 to 8. The connections are as follows:

- Pin 1: +3.3V
- Pin 2: SWD_SWDIO
- Pin 3: SWD_SWDCLK
- Pin 4: SWD_NRESET
- Pin 5: USART2_TX
- Pin 6: USART2_RX
- Pin 7: D8
- Pin 8: GND

The SWD signals (SWD_SWDIO, SWD_SWDCLK, SWD_NRESET) are connected to the SWD pins (1, 2, 3) of the connector. The USART2 signals (USART2_TX, USART2_RX) are connected to the USART2 pins (5, 6) of the connector. The D8 signal is connected to the D8 pin (7) of the connector. The GND signal is connected to the GND pin (8) of the connector.

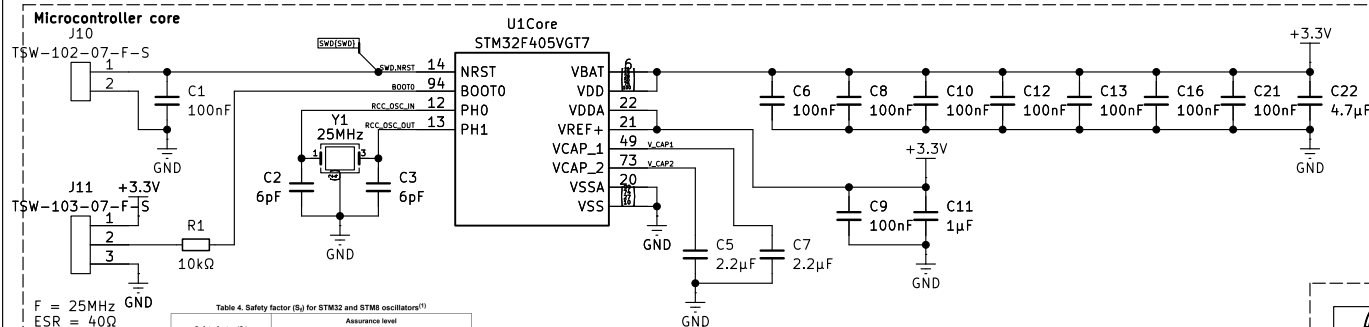


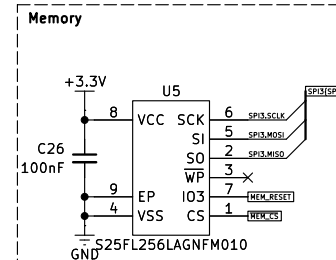
Fig. 2. Safety factor (S_f) for STW32 and STW1 oscillators⁽¹⁾

$f_s = 25\text{MHz}$ GND
 $ESR = 40\Omega$
 $C_L = 8\text{pF}$
 $C_0 = 7\text{pF}$
 $C_S = 5\text{pF}$
 $g_m = 5\text{mA/V}$
 $C_{\text{tot}} = C_L + (C_S / 2)$

Safety factor (S _f)	Assurance level	
	HSE	LSE
$S_f \geq 5$	Safe	Very safe
$3 \leq S_f < 5$		Safe
$S_f < 3$	Not safe	Not safe

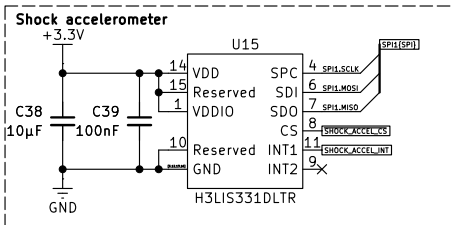
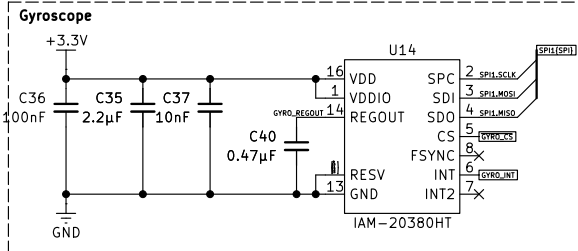
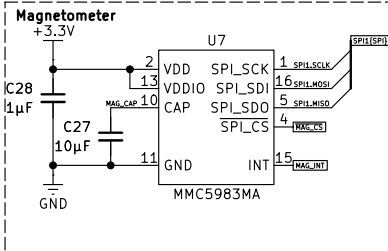
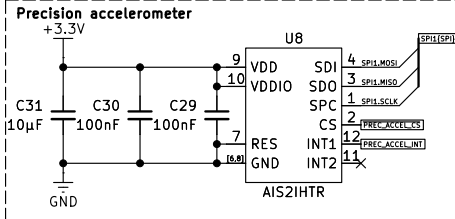
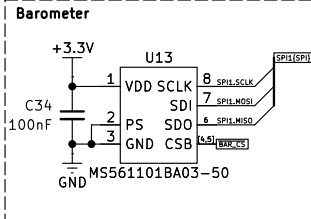
Safe and very safe oscillations are shown in green, unsafe oscillation in yellow.

$C_{L1/2} = 2 \times (C_L - C_S) = 6\text{pF}$
 $CLR = (ESR \times (\pi \times F \times C_{L1/2})^2 \times (V_{pp2})^2) / 2 = 97.07\mu\text{W}$
 $g_m \text{ critical} = 4 \times (ESR) \times (2 \times \pi \times F)^2 \times (C_0 + C_L)^2 = 0.89\text{mA/V}$
 $g_m \text{ margin} = g_m \text{ margin} / g_m \text{ critical} = 5.63$



open source
hardware

Revision:
Sheet 3 of 5



Organization: Lapin Rocketry
 Designer: Gianluca Gaiardi
 Reviewer: Gianluca Gaiardi
 License: CERN-OHL-S
 URL:

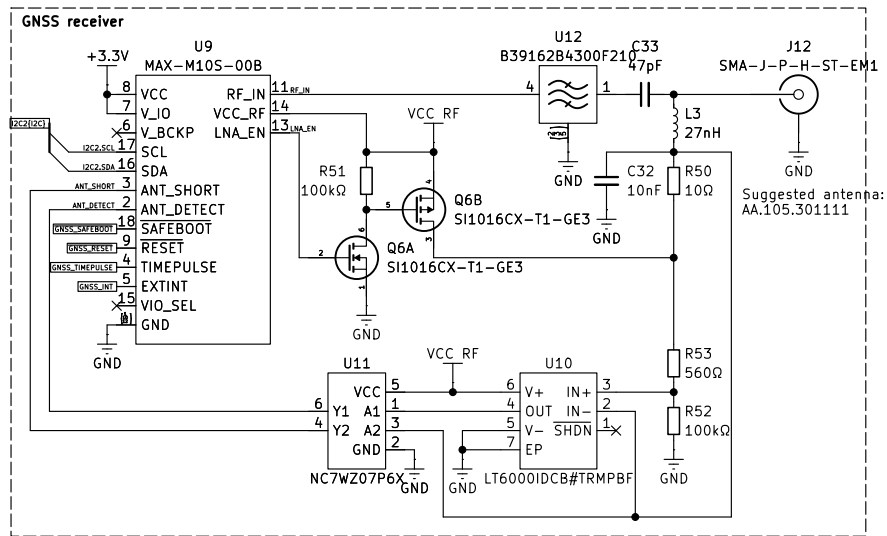


Sheet: /Sensors/
 File: sensors.kicad_sch

Size: A4
 KiCad E.D.A. 10.0.4

Date:

Revision:
 Sheet 4 of 5



Radio
CC1200
CC1201



Organization: Lapin Rocketry
Designer: Gianluca Gaiardi
Reviewer: Gianluca Gaiardi
License: CERN-OHL-S
URL:



Sheet: /RF/
File: rf.kicad_sch

Size: A4
KiCad E.D.A. 10.0.4

Date:

Revision:
Sheet 5 of 5